

For an arbitrary ^{compact} complex manifold

(e.g. proper, non projective variety)

we ask if there is some Hodge decomposition notion.

We can place a guess: the putative Hodge Filtration.

For Kahler manifolds, this will precisely be:

$$F^p H^k(X, \mathbb{C}) = \bigoplus_{r \geq p} H^{r, k-r}(X).$$

We have double complex via Dolbeault resolutions

$$\begin{array}{ccccccc}
 & 0 & & 0 & & & \\
 & \uparrow & & \uparrow & & & \\
 0 \rightarrow & \mathcal{E}_X^{0,n} & \xrightarrow{\partial} & \mathcal{E}_X^{1,n} & \rightarrow & \dots & \rightarrow \mathcal{E}_X^{n,n} \rightarrow 0 \\
 & \uparrow & & \uparrow & & & \uparrow \\
 & \vdots & & \vdots & & & \vdots \\
 & \uparrow & & \uparrow & & & \uparrow \\
 0 \rightarrow & \mathcal{E}_X^{0,1} & \rightarrow & \dots & \rightarrow & \mathcal{E}_X^{n,1} & \rightarrow 0 \\
 & \uparrow \partial & & \uparrow & & \uparrow & \\
 0 \rightarrow & \mathcal{E}_X^{0,0} & \rightarrow & \mathcal{E}_X^{1,0} & \rightarrow & \dots & \rightarrow \mathcal{E}_X^{n,0} \rightarrow 0 \\
 & \uparrow & & \uparrow & & & \uparrow \\
 0 \rightarrow & \mathcal{O}_X & \rightarrow & \Omega_X & \rightarrow & \Omega_X^2 & \rightarrow \dots \rightarrow \Omega_X^n \rightarrow 0 \\
 & \uparrow & & \uparrow & & \uparrow & \\
 & 0 & & 0 & & 0 &
 \end{array}$$

$\mathcal{E}^{i,j} = C^\infty$ sections of $\Omega^{0,j} \otimes \Omega_X^i$

Total complex (once you remove bottom row) is deRham complex

Spectral sequence gives $E_{p,q}^1 = H^q(X, \Omega_X^p) \Rightarrow H_{dR}^{p+q}(X, \mathbb{C})$

Get a filtration on this double complex:

$$\begin{array}{ccccccc}
 & 0 & & 0 & & & \\
 & \uparrow & & \uparrow & & & \\
 & \mathcal{E}_X^{p,n} & \rightarrow & \dots & \rightarrow & \mathcal{E}_X^{n,n} & \rightarrow \dots \\
 & \uparrow & & \vdots & & \uparrow & \\
 & \vdots & & \vdots & & \vdots & \\
 & \uparrow & & \vdots & & \uparrow & \\
 & \mathcal{E}_X^{p,0} & \rightarrow & \mathcal{E}_X^{p+1,0} & \rightarrow & \dots & \rightarrow \mathcal{E}_X^{n,0} \\
 & \uparrow & & \uparrow & & \uparrow & \\
 0 \rightarrow & 0 & \rightarrow & \dots & \rightarrow & 0 \rightarrow \Omega_X^p & \rightarrow \Omega_X^{p+1} \rightarrow \dots \rightarrow \Omega_X^n \rightarrow 0
 \end{array}$$

We get from this filtration

$$F^p H_{\mathbb{R}}^k = \text{Im} (H^k(X, \sigma^{2p} \Omega^\bullet) \rightarrow H^k(X, \Omega^\bullet))$$

The Hodge subspaces are

$$H^{p,q}(X) = F^p H^{p+q}(X, \mathbb{C}) \cap \overline{F^q H^{p+q}(X, \mathbb{C})}$$

If X is Kähler, then

$$\sum_{p+q=k} \dim E_1^{p,q} = \sum_{p+q=k} \dim H^{p,q} = \dim H^k(X, \mathbb{C}) = \sum_{p+q=k} \dim E_\infty^{p,q}$$

Thus, the spectral sequence degenerates at E_1 (as E_{r+1} is a subquotient of E_r , apparently)

Thus, d_p is injective & one can check the images are indeed the

$$\bigoplus_{p \geq r} H^p(X, \Omega^q)$$

Def X a compact complex manifold.

$H^k(X, \mathbb{C})$ has a Hodge decomposition in the strong sense if

1) $\forall p+q=k$, $H^{p,q}(X)$ can be identified w/ the subspace of H^k consisting of classes representable by (p,q) -forms.

& $H^{p,q}(X) \rightarrow H_{\bar{\partial}}^{p,q}(X)$ is an iso.

$$2) \quad H^k = \bigoplus_{p+q=k} H^{p,q}(X)$$

$$3) \quad H_{BC}^{p,q} = \frac{\text{d-closed type } (p,q)}{\partial \bar{\partial} \Gamma E_X^{p-1, q-1}} \rightarrow H_{dR}^{p+q} \otimes \mathbb{C}$$

is injective.

From our def of filtration, we have: if ss degenerates @ $E1$:

$$\text{Prop} \quad H^q(X, \Omega_X^p) \cong \text{Gr}_F^p H_{dR}^{p+q}(X) \quad \text{so}$$

$$\dim H^k(X, \mathbb{C}) = \sum_{p+q=k} \dim H^q(X, \Omega_X^p).$$